DLoc Lightweight WiFi Localization

Data Splits and Experiments Run on the Vast.ai H100

E-JUST graduation project — internal technical record

Compiled October 15, 2026

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1 Overview and scope

This document is a precise, self-contained record of the *data* behind the project: the splits as they are actually defined in the original WILD split-index files, how they are assembled into the three experimental protocols, and every GPU experiment run on the rented Vast.ai H100, annotated with *which split/data it used* and *what the run was for*.

Dataset. WILD (UC San Diego), the public dataset released with DLoc, collected in *Jacobs Hall* with 4 access points (APs). Each labelled example is a $4 \times 161 \times 361$ stack of AoA–ToF heatmaps (one 161×361 image per AP) and a $1 \times 161 \times 361$ Gaussian ground-truth location heatmap on a grid of 0.1 m / pixel (a 16.1 m $\times$ 36.1 m floor).

Error metric. Localization error $e = 0.1 \cdot \|\arg\max(\hat{y}) - \arg\max(y)\|_2$ metres — the Euclidean distance between the peaks of the predicted and ground-truth heatmaps, scaled to metres. Because peaks are integer pixels, e is quantized to the lattice $0.1\sqrt{n}$; medians can coincide across models, so we also report the 90th and 99th percentiles (P90, P99) and, where available, the mean.

Hardware. All GPU training for the project was performed on Vast.ai H100 80 GB instances. The flagship, fully cost-instrumented run (the official leave-one-session-out comparison, Section 6) ran on H100 instance `41279674`; the Phase-1/Phase-2 training (Section 7) ran on the project's earlier Vast.ai H100 instance.

2 The five Jacobs Hall sessions

WILD/Jacobs Hall consists of five data-collection sessions of increasing difficulty. Sessions 1–2 (July 28, 2019) are the "standard" setup used for in-distribution training/testing; Sessions 3–5 (Aug. 16, 2019) introduce distribution shift (furniture, then a specular aluminium reflector) and serve as unseen cross-session test environments.

Roughly 94% of the data is non-line-of-sight (NLOS): field-of-view (FOV) samples are rare (e.g. only 840 of 14,199 in Session 2), so the task is dominated by the hard multipath case.

3 The original split-index files

The splits are *not* ours — they ship with WILD as `data_split_ids_*.mat` files (authored by the DLoc team, dated 2022-09-19). Each file partitions one session's samples into four disjoint

Table 1: The five sessions. "Samples (processed)" is the sample count of the processed dataset file actually used; for Session 1 this is its *training* partition only (see Section 3 and Note 1).

#	Session tag	Date	Samples (processed)	Condition / role
1	`jacobs_July28`	Jul 28, 2019	7,635	reference; training
2	`jacobs_July28_2`	Jul 28, 2019 (+1h)	14,199	same setup; in-dist train+test
3	`jacobs_Aug16_1`	Aug 16, 2019	11,201	furniture rearranged; cross-session
4	`jacobs_Aug16_3`	Aug 16, 2019	8,754	different furniture; cross-session
5	`jacobs_Aug16_4_ref`	Aug 16, 2019	5,486	aluminium reflector; cross-session
		Total processed	**47,275**	

index sets: `fov_train_idx`, `fov_test_idx`, `non_fov_train_idx`, `non_fov_test_idx`. Table 2 reports the exact contents, read directly from the `.mat` files.

Table 2: Exact structure of every original split-index file. "Train" = FOV-train + non-FOV-train; "Test" = FOV-test + non-FOV-test. "Adj." is the fraction of test indices with a train index at position ± 1 — a near-zero value confirms a *contiguous temporal* hold-out (train early, test late), not a random shuffle. All partitions are disjoint and cover $1 \ldots N$ with no gaps.

Split file (`data_split_ids_...`)	FOV-tr	FOV-te	nFOV-tr	nFOV-te	Train	Test	Adj.
Jacobs Hall — used in this project							
`jacobs_July28`	632	271	7,003	3,001	7,635	3,272	0.15%
`jacobs_July28_2`	588	252	9,351	4,008	9,939	4,260	0.07%
`jacobs_Aug16_1`	1,121	480	6,720	2,880	7,841	3,360	0.09%
`jacobs_Aug16_3`	1,001	429	5,127	2,197	6,128	2,626	0.34%
`jacobs_Aug16_4_ref`	342	146	3,499	1,499	3,841	1,645	0.18%
Jacobs Hall — spatial-disjoint (Fig. 13b), NOT run							
`jacobs_July28_space_gen`		(single train/test split)			10,004	903	0.78%
`jacobs_July28_2_space_gen`		(single train/test split)			13,359	840	1.19%
Other sessions — present but NOT used (no feature data locally)							
`July16`	3,417	1,465	14,864	6,371	18,281	7,836	0.17%
`July18`	2,826	1,211	13,828	5,926	16,654	7,137	0.08%
`July22_2_ref`	844	362	3,697	1,584	4,541	1,946	0.26%

Reading the split. Taking `jacobs_July28_2` as the worked example: `non_fov_train_idx` covers indices $1 \ldots 9824$ and `non_fov_test_idx` covers $9825 \ldots 14199$; `fov_train_idx` covers $3728 \ldots 10153$ and `fov_test_idx` covers $10154 \ldots 14016$. Training indices are concentrated in the early part of the recording and test indices in the late part; only 3 of 4,260 test samples (0.07%) have a training neighbour at ± 1. This is a **temporal hold-out**, which matters for defence: with 50 ms sampling from a slow (15 cm/s) robot, a *random* split would leak near-duplicate frames across train/test — but this split does not.

4 From split files to experimental protocols

The split files above are assembled into three protocols. Which processed `.mat` files feed each is fixed in `code/training/train_unet_crosssession.py` (fold table) and `code/training/train_distillation.py` (in-distribution paths).

4.1 (A) In-distribution — DLoc Fig. 10b

Train and test on the two July sessions' *own* partitions.

- **Train (17,574):** `jacobs_July28` training partition (7,635) + `jacobs_July28_2` training partition (9,939 = 588 FOV + 9,351 non-FOV).
- **Test (4,260):** `jacobs_July28_2` test partition (252 FOV + 4,008 non-FOV).

This reproduces the published DLoc training-set size of 17,574 exactly. Session 1's own test partition (3,272 samples) is *not* used by Fig. 10b.

4.2 (B) Zero-shot cross-session — our construction

Train *only* on the two July sessions (the very same 17,574-sample in-distribution training set, i.e. the Phase-1/2 checkpoints are reused), then evaluate, with no fine-tuning, on each *entire* unseen August session.

- **Train:** July28 + July28_2 training partitions (17,574).
- **Test (full sessions):** Aug16_1 (11,201), Aug16_3 (8,754), Aug16_4_ref (5,486).

Purpose: an internal ablation of architecture/distillation under strong distribution shift. It is *not* a fair comparison to DLoc (DLoc trains on three sessions; this trains on two), which is exactly why protocol (C) was run.

4.3 (C) Leave-one-session-out (LOSO) — DLoc's official protocol

The scheme used by the DLoc paper's Table 1: each fold trains on the two July sessions *plus* two of the three August sessions and tests on the held-out August session (used in full). The held-out session never appears in its fold's training set. The August sessions enter training as *full* sessions (no FOV split applied), while the two July sessions contribute their training partitions (17,574, identical to protocol A).

Table 3: The three LOSO folds and their exact sample counts. "J" = the 17,574 July training samples (common to all folds).

Fold	Trains on	Train N	Tests on (held out)
`rw_to_rw_env2`	J + Aug16_3 + Aug16_4_ref	31,814	**Aug16_1** (furniture), 11,201
`rw_to_rw_env3`	J + Aug16_1 + Aug16_4_ref	34,261	**Aug16_3** (diff. furn.), 8,754
`rw_to_rw_env4`	J + Aug16_1 + Aug16_3	37,529	**Aug16_4_ref** (reflector), 5,486

31,814 = 17,574+8,754+5,486; 34,261 = 17,574+11,201+5,486; 37,529 = 17,574+11,201+8,754 — verified against the run logs.

5 Experiments run on the Vast.ai H100 — master list

Table 4 lists every GPU run, the protocol/data it used, and its purpose. Detailed per-run numbers follow in Sections 6–7.

† Robustness applies corruptions (AP dropout, Gaussian noise $\sigma \in \{0.1, 0.2\}$, attenuation, blur) to the protocol-A test set *at test time only* — no retraining.

Table 4: All experiments. "Split" refers to the protocols of Section 4. Results are median localization error unless noted.

Run (model / variant)	Split	Ep. / seeds	Purpose & headline result
DLoc baseline (teacher)	A	50	reference & KD teacher; 0.707 m
DLoc baseline multi-seed	A	120, ×3	seed-stability of DLoc; 0.875 ± 0.036 m
MobileNetV2 (standalone / distilled)	A	50	7.2× compression; 0.707 m (matches DLoc repro)
TinyCNN (standalone / distilled)	A	50	capacity floor probe; 3.92 / 3.71 m (under-capacity)
Mamba	A	50	sequence-model probe; 10.56 m (arch. mismatch)
MobileNetV2-UNet, cosine, standalone	A	120	**main model**; **0.640 m**
MobileNetV2-UNet, step, standalone	A	120	scheduler ablation; 0.640 m
MobileNetV2-UNet, cosine, distilled	A	120	KD on best model; 0.640 m (best tail 2.587 m P99)
Robustness eval (6 models)	A†	—	test-time corruption; KD 3.4× & skips 5.5× noise
Zero-shot cross-session (6 models)	B	reuse	shift ablation; U-Net distilled best (1.020/1.005/1.281)
Official LOSO, U-Net standalone	C	120, ×3	**fair vs published DLoc**; mean 0.930 m vs 0.86

6 Detailed: the official LOSO H100 run

This is the flagship, fully-instrumented H100 run (2026-06-17 → 2026-06-18). Its purpose: re-run *our* MobileNetV2-UNet under DLoc's own leave-one-session-out protocol (protocol C) so the comparison with the published DLoc Table 1 is apples-to-apples (same sessions, same held-out test, same metric in metres), replacing an earlier unfair 2-session cross-session experiment.

Instance & environment. Vast.ai H100 80 GB SXM (instance `41279674`), 192-core EPYC, ∼2 TB RAM, 150 GB NVMe; Python environment `/venv/main`, torch 2.12.0+cu130.

Model. MobileNetV2-UNet (`student_unet.py`): ImageNet-pretrained MobileNetV2 encoder (first conv adapted to 4 AP channels) split into 4 stages with U-Net skip connections into the decoder (320+96→128, 128+32→64, 64+24→32, 32→1 + Sigmoid). **2,409,217 parameters (2.41 M)**, output 161 × 361.

Training recipe (each fold, each seed). Adam, lr = 10^{-4}; cosine annealing ($T_{\max} = 120$); MSE loss; **standalone** (no distillation — see caveat); 120 epochs; batch size 64; seeds 42, 123, 777; best checkpoint = epoch of minimum test median error.

Reading the result. At 6.8× fewer parameters the U-Net *matches* DLoc on the different-furniture fold (0.822 vs 0.82), is within ∼0.07–0.14 m on the others, is level on mean P90 (2.374 vs 2.33), and *beats* DLoc's P90 on the hardest (reflector) fold (2.489 vs 2.77). Medians are seed-

Table 5: All 9 LOSO runs (3 folds × 3 seeds), from the per-run result JSONs.

Fold (held out)	Seed	Median (m)	P90 (m)	P99 (m)	Best ep.
env2 — Aug16_1 (furniture)	42	0.854	1.769	5.445	58
	123	0.806	1.844	5.022	53
	777	0.894	1.903	5.124	40
env3 — Aug16_3 (diff. furn.)	42	0.825	2.640	8.896	67
	123	0.781	2.915	9.986	40
	777	0.860	2.823	9.487	72
env4 — Aug16_4_ref (reflector)	42	1.131	2.474	6.508	52
	123	1.100	2.524	7.101	58
	777	1.118	2.470	6.894	63

Table 6: LOSO aggregated (mean ± sample std, $n = 3$) vs. published DLoc Table 1. Our model is 2.41 M params vs. DLoc's 16.5 M (6.8× smaller).

Fold	Median (ours)	Median (paper)	P90 (ours)	P90 (paper)	P99 (ours)
env2 furniture	$\mathbf{0.852} \pm 0.044$	0.71	1.839 ± 0.067	1.71	5.197 ± 0.221
env3 diff. furn.	$\mathbf{0.822} \pm 0.040$	0.82	2.793 ± 0.140	2.52	9.457 ± 0.546
env4 reflector	$\mathbf{1.116} \pm 0.016$	1.05	2.489 ± 0.030	2.77	6.834 ± 0.301
mean	**0.930**	0.86	**2.374**	2.33	—

stable (±0.016–0.044 m). The residual median gap on the reflector fold is expected: DLoc's consistency (offset) decoder enforces multi-AP agreement under specular multipath, which the single-decoder U-Net lacks.

Compute & cost. ∼40–64 s/epoch; ∼1.5–2 h/run; all 9 runs × 120 epochs completed; wall-clock ≈16 h; cost ≈ \$34–38 at ∼ \$2.16/hr. The DLoc baseline was *not* retrained (by instruction), saving ∼50–100 H100-hours.

Caveats (honest).

1. "Official training strategy" was read as the *session/data protocol*, keeping the U-Net's own recipe (120-ep cosine Adam), not DLoc's optimizer (lr 10^{-5}, batch 32, ∼50 ep).
2. **Standalone, not distilled**: per-fold distillation would need a teacher retrained per fold (excluded by instruction); standalone U-Net is the conservative fair-comparison choice.
3. Comparison is to DLoc's *published* numbers, not our in-house reproduction (which lands ∼0.707 m in-distribution vs the paper's 0.64).
4. Within Jacobs Hall only — not cross-*building* (Atkinson/Fig. 10a data unavailable).
5. The instance had no persistent volume — all 9 run dirs + result JSONs were pulled to local and pushed to HuggingFace before teardown.

7 In-distribution and zero-shot results (Phase 1/2)

These runs trained the teacher, the four students, and the U-Net variants on protocol A (Vast.ai H100, Jun 8–9 2026 per run-dir timestamps), then reused the checkpoints for the robustness

and zero-shot evaluations.

Table 7: In-distribution accuracy (protocol A; Session-2 test set, $N = 4{,}260$). Best in bold. Published DLoc from the paper; our reproduction is the KD teacher.

Model	Params	Mode	Median (m)	P90 (m)	P99 (m)
DLoc (published)	16.5M	—	**0.64**	1.6	3.2
DLoc (our repro)	16.5M	—	0.707	∼1.6	∼4.0
MobileNetV2	2.27M	standalone	0.707	1.616	3.992
MobileNetV2	2.27M	distilled	0.707	1.612	4.003
TinyCNN	47K	standalone	3.922	13.919	23.244
TinyCNN	47K	distilled	3.706	14.784	25.101
MobileNetV2-UNet	2.41M	standalone (cos)	**0.640**	1.523	2.750
MobileNetV2-UNet	2.41M	standalone (step)	**0.640**	1.552	2.918
MobileNetV2-UNet	2.41M	distilled (cos)	**0.640**	**1.513**	**2.587**
Mamba	653K	standalone	10.56	20.0	28.3

Table 8: Zero-shot cross-session (protocol B): train July-only, test full August sessions. Cells: median / P90 (m). Best per column in bold. S2 is the in-distribution reference.

Model	Mode	S2 (in-dist)	Aug16_1	Aug16_3	Aug16_4_ref
MobileNetV2	standalone	0.707	1.118 / 3.138	1.204 / 3.471	1.500 / 3.367
MobileNetV2	distilled	0.707	1.118 / 3.228	1.204 / 3.342	1.442 / 3.711
TinyCNN	standalone	3.922	5.964	6.327	6.605
MobileNetV2-UNet	standalone	0.640	1.063 / 2.508	1.005 / 3.704	1.334 / 3.301
MobileNetV2-UNet	distilled	0.640	**1.020 / 2.319**	**1.005 / 2.608**	**1.281 / 3.002**

The U-Net's 0.640 m is 6.5 standard deviations below this baseline mean, i.e. the improvement is not seed noise. (Note the 120-epoch baseline is *worse* than the 50-epoch teacher because StepLR $\gamma = 0.9$ never decays the learning rate to zero over 120 epochs.)

8 Hyperparameters per model

Verified against `code/` source.

‡ DLoc counts epochs as niter(20) + niter_decay(100) = 120; the KD teacher is a 50-epoch run. StepLR step sizes: encoder 50, decoder 20, offset 50. **Note:** the in-distribution students/U-Net used batch **32**; batch 64 applies only to the baseline and the LOSO U-Net.

9 Notes, de-scoped runs, reproducibility

1. **Session 1 is a training-only partition.** The split file `jacobs_July28` defines 10,907 samples (7,635 train / 3,272 test), but the processed `dataset_jacobs_July28.mat` contains only the 7,635-sample *training* partition — which is why protocol A's training set sums to the published 17,574. Session 1's 3,272 test samples are unused.

Table 9: DLoc baseline multi-seed (protocol A, 120 epochs, StepLR $\gamma = 0.9$).

Seed	Median (m)	P90 (m)	P99 (m)
42	0.825	2.402	22.672
123	0.906	2.335	7.714
777	0.894	2.102	8.694
mean ± std	$\mathbf{0.875 \pm 0.036}$	2.280 ± 0.128	13.027 ± 6.832

Table 10: Training hyperparameters. KD loss (distilled modes): $L = 0.7\,\mathrm{MSE}(\hat{y}, y) + 0.3\,\mathrm{MSE}(\hat{y}, y_T)$, with teacher heatmaps y_T pre-computed offline (the teacher never runs during student training). β=0 recovers standalone.

Model / run	Optimizer	LR	Batch	Epochs	Scheduler / seeds
DLoc baseline	Adam (β_1=0.5)	1e−5	64	120‡	StepLR γ=0.9
DLoc multi-seed	Adam (β_1=0.5)	1e−5	64	120	StepLR; seeds 42/123/777
MobileNetV2, TinyCNN	Adam	1e−4	32	50	—; seed 0
Mamba	Adam	1e−4	32	50	—; seed 0
U-Net (in-dist)	Adam	1e−4	32	120	cosine $T_{\max}$=120 *or* step(20, 0.5)
U-Net (LOSO)	Adam	1e−4	64	120	cosine $T_{\max}$=120; seeds 42/123/777

2. **August split files are unused.** In protocols B and C the August sessions are always used in *full* (no FOV split), so the `data_split_ids_jacobs_Aug16_*` files (Table 2) do not enter any run.

3. **Present but never run:** the spatial-disjoint `*_space_gen` splits (DLoc Fig. 13b); the `July16`/`July18`/ `July22_2_ref` sessions (a different building / setup — no feature data held locally); Fig. 10a (Atkinson, 3 APs) and Fig. 13a (20/40 MHz bandwidth).

4. **De-scoped (budget):** student multi-seed (MobileNet/TinyCNN/U-Net ×2 seeds); per-fold distilled LOSO. Two further studies (Study A: spatial-preservation vs. parameter count; Study C: distillation mechanism) are built but not yet run.

5. **Reproducibility.** Code: GitHub `Abdulrahman-bits2/DLoc-lightweight-localization`. Processed dataset (70.5 GB, all 5 sessions) + split-index files: HuggingFace `JustAGeek/dloc-wild-fig10b`. Trained weights, per-epoch logs, result files, and the 9 LOSO run dirs: HuggingFace `JustAGeek/dloc-code` (`weights/`, `logs/`, `results/`, `crosssession_official/`). Teacher heatmaps regenerate via `precompute_teacher.py`.